ATTENDANCE MODULE · COMPLETE GUIDE

# Attendance Engine — Complete Guide

How the engine works end to end, what HR can configure, which rule wins when two rules could apply to the same day, and a real month-long walkthrough showing the rules in action.

Audience: HR, product owners, and client stakeholders. No database, API, or code knowledge is assumed.

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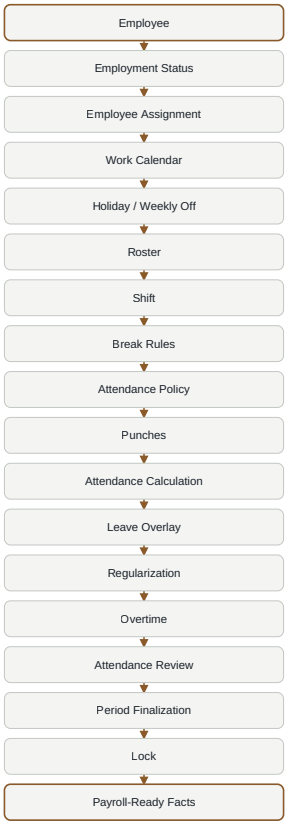
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PART 1

# How the engine works

Every employee, on every date, is run through the same question in three parts: was the employee expected to work (calendar, holidays, weekly-off, roster) — what schedule were they expected to follow (shift, breaks) — and based on what actually happened, what's the final result (punches, judged against the Attendance Policy, then adjusted for approved leave, corrections, and overtime).

## The end-to-end flow




## What each stage decides

### Employment status

Was the employee actually employed on this date?

**Controls:** HR (joining/exit dates) · **Overridden by:** nothing — checked first, absolute

### Employee assignment

Branch, department, designation, grade, manager, default shift — as of that date.

**Controls:** HR · **Overridden by:** a published roster can replace the shift for scheduling

### Work calendar

Which holiday list and weekly-off pattern apply.

**Controls:** HR, via calendar assignment · **Overridden by:** the most specific assignment (Part 3.1)

### Holiday / weekly-off

What kind of day is it — working, holiday, or weekly off.

**Controls:** calendar + weekly-off pattern · **Overridden by:** a date-specific exception, or one documented carve-out (Part 3.2)

### Roster

Has this employee been scheduled differently for this specific date?

**Controls:** HR/manager with roster access · Must be **published** to count; never changes the calendar's day-type

### Shift

Actual work timings and hours expected.

**Controls:** HR · **Overridden by:** roster, for scheduling on a given date

### Break rules

What breaks are scheduled and how they affect worked hours.

**Controls:** HR, on the shift · Only breaks matching an actual punch gap are deducted

### Attendance policy

How lateness, absence, half-days, and missing punches are judged.

**Controls:** HR · **Overridden by:** the most specific policy assignment (Part 3.1)

### Punches

What the employee actually recorded.

**Controls:** employee, device, HR/manager, or a regularization · Never edited or deleted, only added to

### Attendance calculation

Turns the day's expectations and punches into Present / Half-Day / Absent / Incomplete.

**Controls:** the engine, automatically · Late/early bands can force a worse status than hours worked alone (Part 3.3)

### Leave overlay

Reflects approved leave onto the day.

**Controls:** Leave system (external), synced in · Wins by default; never changes the calendar's day-type

### Regularization

The correction workflow for a wrong or missing punch.

**Controls:** employee submits, manager/HR approves · Adds new punches, day recalculated from scratch

### Overtime

Detects, classifies, and routes extra worked time for approval.

**Controls:** policy + approver · Never changes Present/Absent/Half-Day — always a separate fact

### Attendance review

HR checks for gaps before closing the period.

**Controls:** HR/Admin

### Period finalization & lock

Freezes the month's results for payroll.

**Controls:** HR/Admin with lock permission · Blocks any change until explicitly reopened

### Payroll-ready facts

Final paid/unpaid fraction and approved OT minutes per employee, per day.

Hands off facts only — does not calculate money

## What kind of day is it?

Every date is classified in this strict order: **1.** a specific date-level exception, if HR created one · **2.** otherwise a holiday, if listed in the employee's holiday calendar · **3.** otherwise weekly off, if it matches the weekly-off pattern · **4.** otherwise a normal working day.

**The one documented exception:** if a public holiday falls on an employee's weekly-off day, the day **stays classified as Weekly Off** — no substitute working day is created automatically. The coincidence is recorded and visible to HR, but nothing is auto-compensated. If the business wants a substitute day off, HR must manually add a Substitute Holiday exception elsewhere. Many Indian HR policies expect an automatic substitute — this needs explicit client confirmation, not a system default to assume.

## The shift system

Shift defines the **timetable** — it does not define how lateness or absence is judged; that is the Attendance Policy's job.

### Fixed

Same start/end every day. Lateness measured against the fixed start; early departure against the fixed end.

### Flexible

Arrive/leave within a window, complete required hours. A "core hours" window can be entered but is not currently enforced.

### Split

Two or more work blocks in a day. Each needs its own punch pair; a missed block's outcome is configurable. Lateness isn't calculated — only whether hours were met.

### Night / cross-midnight

Starts one day, ends after midnight. The entire shift is attributed to the date it started — never split.

## Break rules

**Active today:** matching a punch gap to a defined break window; deducting unpaid break time; adding paid break time back; flagging breaks longer than the configured maximum.

**Not active today:** a "mandatory" break is not force-deducted without a matching punch gap; minimum-duration limits; any automatic consequence for exceeding maximum break duration beyond recording the excess.

PART 2

# What's configurable

Every number and rule below is set by HR — nothing is hard-coded. Each policy area can be assigned at different scopes (see Part 3), so different teams can run under entirely different rules.

## Work calendar & weekly off

| SETTING | EXAMPLE | ACTIVE |
|---|---|---|
| Calendar identity, holiday list composition | Ordered list of holiday calendars | Yes |
| "Optional holidays are hard holidays" flag | On/Off, company-wide | Yes |
| Calendar exceptions | Shutdown, Special Working Day, Substitute Holiday, Cancel Holiday | Yes highest priority |
| Weekly-off pattern | 1st/3rd Saturday, alternate Saturdays, etc. | Yes all patterns |

## Shift & break

| SETTING | EXAMPLE | ACTIVE |
|---|---|---|
| Shift type, timing, required minutes | Fixed/Flexible/Split, 09:00–18:00, 480 min | Yes |
| Cross-midnight / OT-eligible / operates-on-holiday flags | On/Off | Yes |
| Break window, paid/unpaid, max duration | 13:00–14:00, unpaid, 60 min cap | Yes |
| Core hours (mandatory presence window) | 11:00–16:00 | No stored only |
| Mandatory break force-deduction | Deduct even with no matching punch gap | No |
| Action on exceeding max break | Flag / other consequence | No recorded as fact only |

## Attendance policy

| SETTING | EXAMPLE | ACTIVE |
|---|---|---|
| Grace-In / Grace-Out minutes | 15 min / 10 min | Yes |
| Late bands (lateness → status) | 30+ min late → Half-Day | Yes overrides hours worked |
| Early-departure bands | 60+ min early → flag | Yes |

| SETTING | EXAMPLE | ACTIVE |
|---|---|---|
| Hour bands (worked minutes → status) | ≥480 Present, 240–479 Half-Day, <240 Absent | Yes |
| Missing-punch behaviour | Incomplete (default) / Absent / Require Correction | Yes "Auto-close" is a naming trap |
| Multi-punch mode, duplicate-punch window | Sum all pairs, 60-sec dedupe | Yes |
| Work-on-leave / Incomplete-at-lock behaviour | Leave wins; treat Incomplete as Absent at lock | Yes |
| Punch rounding, core-hours breach behaviour | Round to nearest 15 min | No |
| WFH / On-Duty as a distinct status | — | No — see note |
| Repeated-lateness-to-half-day conversion | "3 lates in 30 days = 1 half-day" | No |

> **WFH / On-Duty:** these exist as regularization request types and can be submitted and approved, but approval alone does not produce a distinct WFH or On-Duty status — only an effect if it also adds punches or changes the shift. Treat as not fully available until confirmed as a requirement.

## Overtime & regularization policy

| SETTING | EXAMPLE | ACTIVE |
|---|---|---|
| Eligibility mode | Never / Shift decides / Always | Yes |
| Eligibility "by grade" | — | No behaves like "Always" |
| Minimum minutes, rounding, daily/weekly/monthly caps | 30 min min, round to 30, 120/600/2400 caps | Yes |
| Approval chain, comp-off instead of pay | Auto / Manager / HR / Manager-then-HR | Yes |
| Rate multipliers (e.g., "2x for holiday") | — | No reference only, never paid |
| Night OT classification | Shift-flag / Window / Both | No default mode never classifies Night |
| Who may request, backdating window, monthly limit | 7-day backdate, 3 requests/month | Yes |
| Blocked when period locked | On (default) | Yes |

PART 3

# What overrides what

## Policy specificity — the most specific level wins

Calendars and the Attendance / Overtime / Regularization policies can each be assigned at eight levels. The system checks from most specific to least specific and stops at the first match:

`EMPLOYEE` → `SHIFT` → `DESIGNATION` → `GRADE` → `DEPARTMENT` → `BRANCH` → `LEGAL ENTITY` → `COMPANY`

Each policy family resolves independently — an employee-level Attendance Policy says nothing about which level of Overtime Policy applies to that same employee.

| SCENARIO | WINNER | WHY |
|---|---|---|
| Company Grace-In = 15 min; nothing more specific set | Company (15 min) | Only level with an assignment |
| Company = 15 min; Department "Sales" = 10 min | Department for Sales; Company elsewhere | Department is more specific |
| Company = 15; Department = 10; Employee = 5 | Employee (5 min) | Employee is most specific |
| Shift "Night A" has its own OT Policy; Dept "Security" has a different one | Shift-level policy | Shift outranks Department |
| No policy assigned at any level | **Configuration Error** | No silent default — fails closed, blocks period close |
| Employee transferred Dept A → B mid-month, different Reg Policies | A before transfer, B after | Resolution is always as of the relevant date |

**A documented mismatch worth knowing:** if a policy is assigned at the Shift level, and an employee is temporarily moved via a published roster, the system still looks up the Shift-level policy using the employee's **originally assigned** shift — not the roster's shift. Open product decision, not a bug — worth a specific sign-off if shift-scoped policies and roster reassignment are both in use.

## Calendar precedence

| PRIORITY | RULE | RESULT |
|---|---|---|
| 1 (highest) | Date-specific exception | Exception's day-type wins outright |
| 2 | Listed holiday | Holiday / soft-holiday, per its type |
| 3 | Weekly-off pattern match | Weekly Off |

| PRIORITY | RULE | RESULT |
|---|---|---|
| 4 (lowest) | Nothing else matched | Working Day |

> Special case: a hard holiday landing on a weekly-off day stays Weekly Off, flagged as a coincidence — the one documented carve-out to this order.

## Attendance status precedence — inside a single day's calculation

1. **Employment gate** — Not Joined / Exited, absolute, no exceptions
2. **Configuration Error** — missing calendar or Attendance Policy
3. **Roster week-off with no punches** → Not Expected
4. **Missing shift** → follows the missing-shift rule
5. **Not-expected day with no punches** → Not Expected
6. **Punch-based evaluation** → Present / Half-Day / Absent / Incomplete
7. **Leave overlay** — runs last, can still change the outcome

**The rule most worth explaining to a client directly:** a late-arrival classification of Half-Day or Absent from step 6 **overrides** whatever the hours-worked bands alone would have said. An employee who arrives 35 minutes late and then stays late to make up the time is still marked Half-Day — punctuality and total hours are judged independently, and the less favorable outcome applies. See the Thursday scenario in Part 4.

## Quick answers — "Can X override Y?"

| | |
|---|---|
| Can an Employee-specific policy override the Company policy? | Yes |
| Can a Shift-specific policy override a Department policy? | Yes |
| Can a Roster entry override the employee's assigned shift? | Yes, for scheduling |
| Does a Roster entry also change which Shift-scoped policy applies? | No |
| Can a Calendar Exception override a Holiday? | Yes |
| Can a Holiday override a Weekly Off? | No — stays Weekly Off, flagged |
| Can Leave override an Absence or a Not-Expected day? | Yes |
| Can Leave override a Holiday's day-type? | No |
| Can a Punch override Leave? | Only in "flag work-on-leave" mode |
| Can Regularization override an original Punch? | Only by adding new punches |
| Can Overtime override the Attendance status? | No — always parallel |
| Can a Period Lock override all mutating actions? | Yes |
| Can a Draft roster override the live schedule? | No — only Published counts |

PART 4

# A day (and a month) in the life

The rules above are abstract until applied to a real month. The scenarios below follow one employee through a representative set of situations, showing exactly what the system determines at each step, and why.

## Employee setup used in this walkthrough

| | |
|---|---|
| Shift | General Shift, 09:00–18:00 (9 hours) |
| Lunch break | 13:00–14:00, unpaid, 1 hour |
| Required work | 480 minutes (8 hours), net of lunch |
| Grace period (late arrival) | 15 minutes — punch-in by 09:15 is on time |
| Grace period (early departure) | 10 minutes — punch-out from 17:50 is on time |
| Late band | Arrival 30+ min late forces Half-Day, regardless of hours worked |
| Hours band | 8+ hrs → Present · 4–8 hrs → Half-Day · under 4 hrs → Absent |
| Weekly off | Sunday |
| Overtime | Requires manager approval; minimum 30 minutes to qualify |

> These figures are configured by HR for this example team — a different team could run entirely different numbers (see Part 2).

MONDAY

IN 08:57

OUT 18:03

### A standard day

Present

On time, full day worked, nothing to flag. No configuration or approval is involved — the default case that resolves silently for most of the workforce on most days.

| RESULT | PAYROLL FACT |
|---|---|
| Present | 1.00 paid day |

TUESDAY

IN 09:10

grace 15m

## Late arrival within grace

Present

10 minutes after the 09:00 start, but within the 15-minute grace period — treated identically to 09:00.

| RESULT | PAYROLL FACT |
|---|---|
| Present, not flagged | 1.00 paid day |

This is the purpose of a grace period: it absorbs ordinary variation in arrival time without anyone needing to notice, flag, or approve anything.

WEDNESDAY

IN 09:25

OUT 18:00

## Late beyond grace, no status impact

Present

Raw lateness 25 min, minus 15-min grace → 10 min chargeable — under the 30-min Half-Day late band, so status is unaffected.

| RESULT | PAYROLL FACT |
|---|---|
| Present, flagged "late" (10 min) | 1.00 paid day — flag informational |

THURSDAY

IN 09:50

OUT 18:45

## Lateness overriding hours worked

Half-Day

Arrives 50 min late, stays back an extra 45 min to compensate. Worth explaining to the client directly:

- Chargeable lateness: 50 − 15 = **35 minutes**, crossing the 30-min late band.
- Hours worked exceed 8 — on hours alone, this would be Present.
- The system checks the **late band first**. Once crossed, the day is forced to Half-Day regardless of hours worked after.

| RESULT | PAYROLL FACT |
|---|---|
| Half-Day, despite extra hours | 0.50 paid, 0.50 unpaid |

**Rationale:** punctuality is judged independently of total hours. Correctable only via an approved regularization request; otherwise the Half-Day stands.

FRIDAY

IN 08:58

OUT 16:40

## Early departure

Present / Half-Day

Punch-out 80 min before the 18:00 end — 70 min past the 10-min grace, crossing the "60+ min early = flag" band.

| RESULT | PAYROLL FACT |
|---|---|
| Flagged — Present or Half-Day, by severity | Matches whichever status the band resolves to |

MON (WK 2)
IN missing
OUT 18:05

## A missed punch

Incomplete → Present

Forgets to punch IN; punches OUT at 18:05. Missing-punch policy defaults to Incomplete, not automatic Absent.

- **Resolution:** submits a regularization request within the 7-day backdating window.
- Manager approves — system adds the missing IN punch and recalculates, typically resolving to Present.

| RESULT AFTER APPROVAL | PAYROLL FACT |
|---|---|
| Present | 1.00 paid day, once approved |

If not submitted before the period locks, the correction is refused until HR reopens the period — timely submission matters more as month-end approaches.

WED (WK 3)
leave AM
IN 14:05
OUT 18:10

## Half-day approved leave

Half-Day · planned

Approved half-day leave AM, punches in the afternoon. Leave credit (0.5) combined with worked hours.

| RESULT | PAYROLL FACT |
|---|---|
| Half-Day — planned, not a penalty | ~1.00 paid day (0.50 leave + 0.50 worked) |

Distinct from Thursday's Half-Day: that one was a lateness penalty; this one is an approved outcome and needs no review.

THU (WK 3)
no punches
sick leave

## Full-day approved leave

Leave

Approved sick leave, no punches. The leave overlay takes precedence over an empty punch record — no gap is evaluated.

| RESULT | PAYROLL FACT |
|---|---|
| Leave — not Absent, not Incomplete | Paid or unpaid per the leave type |

SUNDAY
IN 09:05
OUT 15:30

## Work on a weekly off

Present OT Pending

Rostered in for a client shipment — 6 hours worked. All hours on a weekly-off day are evaluated as overtime, classified "Weekly-Off overtime," and held Pending until manager approval.

| RESULT | PAYROLL FACT |
|---|---|
| Present + Pending Overtime (6 hrs) | 1.00 paid day + OT once approved |

If still Pending at month-end, period close is **blocked** until decided — the system neither drops nor auto-approves it.

## Month-end — locking a payroll-ready result

1. **Every pending regularization** must be decided.
2. **Every pending overtime request** must be decided.
3. Any day still **Incomplete** is resolved per the "incomplete-at-lock" setting.
4. Once resolved, HR **locks the period** .

After the lock: no new punches, leave syncs, regularizations, or overtime requests can affect that period. A correction needed after lock requires HR to explicitly reopen it first.

| DAY | STATUS | PAID | UNPAID |
|---|---|---|---|
| Mon | Present | `1.00` | `0.00` |
| Tue | Present (grace absorbed) | `1.00` | `0.00` |
| Wed | Present (late flagged, no cost) | `1.00` | `0.00` |
| Thu | Half-Day (late band, despite extra hours) | `0.50` | `0.50` |
| Fri | Present / Half-Day (early-departure band) | `per band` | `per band` |
| Mon (wk 2) | Present (missed punch, regularized) | `1.00` | `0.00` |
| Wed (wk 3) | Half-Day (planned leave) | `~1.00` | `~0.00` |
| Thu (wk 3) | Leave (full day) | `per leave type` | `per leave type` |
| Sun | Present + OT (rostered) | `1.00 + OT` | `0.00` |

> Every row is traceable to a specific punch, leave record, or approval — this is the figure payroll actually consumes.

PART 5

# Summary reference

| SITUATION | GOVERNING RULE | TYPICAL RESULT |
|---|---|---|
| Arrival within grace period | Grace-in minutes | Present, not flagged |
| Late arrival beyond grace, under the Half-Day late band | Late band (lower tier) | Present, flagged late |
| Late arrival crossing the Half-Day late band | Late band (upper tier) — overrides hours worked | Half-Day, regardless of hours worked |
| Early departure within grace | Grace-out minutes | Present, not flagged |
| Early departure beyond grace and the early band | Early-departure band | Present-flagged or Half-Day, by severity |
| Missed punch | Missing-punch policy (default: Incomplete) | Incomplete, resolved via regularization |
| Half-day approved leave, worked the remaining half | Leave overlay + worked hours | Half-Day (planned, not a penalty) |
| Full-day approved leave, no punches | Leave overlay takes precedence | Leave |
| Work on a weekly off or holiday | Roster + overtime detection | Present, plus Overtime (Pending until approved) |
| No policy assigned at any level | Fails closed | Configuration Error — blocks period close |
| Outstanding item at month-end | Period finalization checks | Blocks lock until resolved |